# Muhammad Mushfiqur **Rahman**

Data Engineer

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Data Engineer with an engineering and computational research background, combining hands-on cloud data engineering with experience developing physics-based computational models for industrial engineering applications. Currently the technical owner of a greenfield Snowflake data platform, leading solution design and delivery across production ELT pipelines, data models and quality frameworks. Experienced in taking complex technical problems from ambiguous requirements through to practical implementation, leading technical delivery and communicating effectively across engineering, analytical and business stakeholders.

## Work Experience

### University of Birmingham | IT Services

*Birmingham, UK*

Lead Data & Analytics Developer

*May. 2025 — Current*

- Lead engineer on a greenfield data platform for a high-profile Dynamics 365 CRM transformation programme, with technical ownership across solution design, delivery, infrastructure, data modelling and quality.
- Lead the solution design and delivery of the data and reporting workstream, taking requirements from discovery through architecture, technical design, implementation and release.
- Coordinate delivery across a small multidisciplinary team comprising a BI Platform Engineer, Data Analyst, Data Engineer contractor and myself, including technical work allocation, engineering standards and technical interviews.
- Designed and built the end-to-end ELT platform ingesting Dynamics 365 CRM data into Snowflake via Azure Synapse Link, resolving complex cloud infrastructure and integration blockers across multiple teams.
- Architected the dbt data layer across raw, bronze, silver and gold, establishing modelling conventions, transformation patterns, documentation and development standards from scratch.
- Designed the long-term data retention, deletion and referential-integrity architecture for CRM data, supporting GDPR deletion requirements without compromising data lineage or analytical integrity.
- Implemented masterid-based merge tracking, cascade-of-deletes processing and SCD Type 2 modelling across key CRM entities, supporting approximately 1 million contact records through deduplication and GDPR purge cycles.
- Developed a Python-based automated data-quality, validation and regression-testing framework integrated into deployment workflows to monitor pipeline accuracy and integrity across data layers.
- Managed the development, test and production lifecycle using Git and CI/CD, with ownership of releases and environment promotion across data platform assets.
- Led stakeholder engagement across programme workstreams, translating business requirements into technical data solutions and acting as a senior technical point of contact across infrastructure, engineering, business and reporting teams.
- Mentored data analysts and supported wider data capability through super-user groups, technical guidance and self-service adoption of the data platform.

### University of Birmingham | IT Services

*Birmingham, UK*

Data & Analytics Developer

*Aug. 2024 — May. 2025*

- Joined the data product team during the early stages of a greenfield Dynamics 365 CRM platform, contributing across data modelling, reporting and pipeline development before progressing to lead engineering responsibility.
- Developed Power BI reporting and data models directly against Dataverse, using Power Query to transform CRM data into operational and strategic reporting capabilities.
- Developed SQL-based ELT pipelines and dbt models for CRM datasets, applying dimensional modelling and data-warehousing principles as the platform evolved towards Snowflake.
- Delivered reporting and data assets through Git-based development workflows and agile delivery methods, contributing to controlled deployment and iterative delivery.
- Led agile ceremonies for the data and reporting function, coordinating sprint planning, prioritisation, stand-ups and delivery discussions across technical and analytical contributors.
- Worked directly with stakeholders to translate reporting and data requirements into technical solutions, bridging operational reporting needs with the underlying data architecture.

### Falaq Wholesaler | Technology & Data

*Birmingham, UK*

Data & Operations Analyst

*Aug. 2020 — Sep. 2024*

- Managed and analysed product, sales and inventory data across wholesale, ecommerce and internal operational systems, supporting commercial reporting and day-to-day decision-making.
- Configured and implemented the company's SalesPresenter wholesale sales application, working with the external technical team at Blue Alligator to translate business requirements into the platform and support adoption by the sales team.
- Managed ecommerce data and product catalogue information through the company's WordPress/WooCommerce webshop, maintaining product, pricing, inventory and order data across online and internal systems.
- Designed and implemented structured SKU, product and inventory tracking processes, improving data consistency, stock visibility and reconciliation across multiple systems.
- Performed reconciliation between internal records, wholesale systems and online databases, identifying and resolving data discrepancies.
- Owned data and operational systems end-to-end in a small business environment, working independently with imperfect data, external technology providers and changing requirements.

**University of Birmingham** | **Rolls-Royce Industrial Collaboration** *Birmingham, UK*

PhD Researcher — Integrated Computational Materials Engineering *2016 — 2020*

- Industrial research programme focused on computational modelling of microstructure evolution in nickel-based superalloys for aerospace turbine-disc applications in collaboration with Rolls-Royce plc.
- Developed in-house physics-based simulation code using FORTRAN and Thermo-Calc to model microstructure evolution and precipitate growth in nickel-based superalloys.
- Implemented computational methods including finite-difference, level-set, linear elasticity, mean-field and multicomponent diffusion approaches to solve complex materials-science problems.
- Developed and validated computational models for industrial use, delivering user-oriented models, documentation and technical presentations to the Rolls-Royce Materials Modelling group.
- Worked across computational modelling and laboratory experimentation, using SEM, TEM and optical microscopy alongside Gleeble thermo-mechanical testing and furnace heat treatments.
- Independently investigated complex technical problems through literature research, computational development and experimental validation while collaborating with academic and industrial specialists.

**University of Birmingham** | **Dept. Of Physics** *Birmingham, UK*

Teaching Associate / Teaching Assistant *2014 — 2018*

- Taught and demonstrated undergraduate Physics laboratory sessions and provided assessment and feedback for second-year courses including Classical Mechanics and Statistical Physics.
- Taught postgraduate English for Academic Purposes, helping students communicate technical knowledge, interpret academic literature and develop research papers.
- Developed extensive experience communicating complex technical concepts clearly, providing structured feedback and supporting learners with varied technical backgrounds.

# Education

**University of Birmingham** *Birmingham, UK*

Data & Analytics *Jul. 2023 - Jan. 2024*

- Intensive training covering Python, SQL, R, machine learning, cloud deployment, big-data technologies, BI platforms and Git-based development workflows.

**University of Birmingham** *Birmingham, UK*

M. Res.-Level Research in Integrated Computational Materials Engineering *Jan. 2014 - Dec. 2016*

**The University of Cambridge** *Cambridge, UK*

M. Eng. Mechanical Engineering *Aug. 2011 - Jul. 2012*

**The University of Cambridge** *Cambridge, UK*

B.A. Mechanical Engineering *Sep. 2008 - Jul. 2011*

# Skills

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| **Data Engineering & Platforms** | Design and delivery of cloud data platforms and production ELT/ETL pipelines<br>Snowflake, Azure, Synapse Link, dbt, SQL, Python, Git |
| **Data Modelling & Quality** | Dimensional modelling, SCD Type 2, data quality, validation and governance<br>Data lineage, documentation, reconciliation, regression testing and GDPR-aware data design |
| **CRM & Analytics Data** | Engineering data solutions around enterprise CRM and reporting platforms<br>Dynamics 365, Dataverse, Power BI, Power Query and CRM data integration |
| **Software Engineering & CI/CD** | Production development using Python and SQL with Git-based engineering workflows<br>Automated testing, CI/CD, code review, version control and reproducible deployments |
| **Architecture & Technical Leadership** | Leading technical solution design from requirements through architecture and delivery<br>Technical standards, resource coordination, mentoring, stakeholder management and technical interviewing |
| **Scientific Computing** | Computational modelling and scientific software development in materials engineering<br>FORTRAN, Thermo-Calc, numerical modelling, materials data analysis and industrial research |